Short Term Electricity Load Forecasting for Smart Meter Data

Synthetic Authors (demo data)

Abstract

Utilities need accurate short term forecasts of household electricity demand to plan supply. We present a gradient boosting forecaster for smart meter data. The model reaches a mean MAPE of 4.2 percent on our test split and beats two named baselines.

1. Introduction

Smart meters report household consumption at regular intervals. Accurate forecasts reduce the cost of balancing supply and demand and help utilities schedule generation. Existing methods are often hard to deploy on large meter fleets.

2. Method

The forecaster uses gradient boosted trees on lagged consumption, hour of day and day of week. Features are computed from a sliding window of past readings. The model is retrained once per week.

3. Experiments

3.1 Setup

We evaluate the forecaster on smart meter data from households. The dataset contains 1,200 households. We compare against two baselines, a seasonal naive forecaster and an LSTM network. All results come from a single run on the test split.

3.2 Results

Our forecaster reaches a MAPE of 4.2 percent on the test split. Baseline MAPE is 7.9 percent for the seasonal naive forecaster and 5.1 percent for the LSTM network. Errors are lowest in the early morning and highest during evening peaks.

4. Conclusion

We presented a gradient boosting forecaster for smart meter data and showed that it forecasts household demand accurately on our data.